



E0361

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Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Operations Research

Exam: IN0024 / Endterm

Date: Friday 7th August, 2026

Examiner: Prof. Dr. Martin Bichler

Time: 17:00 – 19:00

Working instructions

- This exam consists of **20 pages** with a total of **7 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 121 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
- Subproblems marked by * can be solved without results of previous subproblems.
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____



☐ Exam empty





Problem 1 Multiple Choice Questions (16 credits)

Mark all correct answers (there can be more than one):

Scoring: Each question is worth up to 2 points. You receive 2 points if you select all correct options and no incorrect ones. Selecting one incorrect option or not selecting a correct option reduces the score to 1 point. Any greater deviation results in 0 points.

Mark correct answers with a cross



To undo a cross, completely fill out the answer option



To re-mark an option, use a human-readable marking



a) Which of the following statements are correct?

- ☐ Consider $\min\{c^T x : Ax \leq b, x \in \mathbb{Z}^n\}$. If A is totally unimodular, then the optimal solution can be found in polynomial time.
- ☐ Every linear program can be solved in polynomial time.
- ☐ The branch-and-bound algorithm is guaranteed to find an optimal solution to an integer program in polynomial time.
- ☐ The simplex algorithm always finds an optimal solution to a linear program in polynomial time.

b) Which of the following statements are correct?

- ☐ If both the primal and dual problems are feasible, then they both have optimal solutions.
- ☐ If the primal problem is unbounded, then the dual problem is infeasible.
- ☐ If the dual problem is unbounded, then the primal problem is infeasible.
- ☐ If the feasible region is unbounded, then any linear program defined on it has an unbounded objective value.

c) Consider a maximization integer linear program solved using the Branch-and-Bound algorithm. Which of the following statements are correct?

- ☐ Branch-and-Bound always explores every feasible integer solution before proving optimality.
- ☐ If the LP relaxation at a node has an integer-optimal solution, then that solution is optimal for the corresponding subproblem.
- ☐ A node can be pruned if the value of its LP relaxation is worse than the value of the current incumbent solution.
- ☐ For a maximization problem, the objective value of the LP relaxation provides an upper bound on the objective value achievable by any integer-feasible solution in that node.

d) Consider a pure integer linear program whose LP relaxation has been solved optimally. The optimal LP solution is fractional. Which of the following statements are correct?

- ☐ If a Gomory cut is added to the LP relaxation, the optimal objective value of the LP relaxation can only improve (increase) for a maximization problem.
- ☐ A Gomory cut can be derived from a row of the optimal simplex tableau whose basic variable has a fractional value.
- ☐ A Gomory cut removes the current fractional LP-optimal solution but does not remove any feasible integer solution.
- ☐ A Gomory cut can be generated from the objective row of the simplex tableau.





e) Which statements about total unimodularity are correct?

- ☐ Every square matrix with determinant 1 is totally unimodular.
- ☐ A matrix containing an entry 2 cannot be totally unimodular.
- ☐ In a totally unimodular matrix, every square submatrix has determinant in $\{-1, 0, 1\}$.
- ☐ If A is totally unimodular and b is integral, then every feasible point of $\{x : Ax \leq b\}$ is integral.

f) A 1.5-approximation algorithm for a metric TSP instance returns a tour of cost 30. Let OPT denote the cost of an optimal tour. Which conclusions are guaranteed?

- ☐ $\text{OPT} = 20$.
- ☐ $\text{OPT} \geq 20$.
- ☐ The returned tour must be optimal.
- ☐ $\text{OPT} \leq 30$.

g) Which of the following statements are correct?

- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, then every stationary point of f is a local maximum.
- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, then every local minimizer of f is also a global minimizer.
- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex, then it always has a minimizer.
- ☐ If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable and $\nabla^2 f(x) \succeq 0$ for all x , then f is convex.

h) Which of the following statements are correct?

- ☐ A KKT point of a non-convex optimization problem is always a global optimum.
- ☐ For a convex minimization problem satisfying Slater's condition, the KKT conditions are necessary for optimality.
- ☐ Slater's condition requires that all feasible points satisfy all inequality constraints strictly.
- ☐ For a convex minimization problem, any feasible point satisfying the KKT conditions is a global minimizer.

Additional space for notes:





Problem 2 Glass Production (21 credits)

A manufacturer develops specialty glasses. The glass is produced by mixing different oxides. The composition influences the physical properties of the glass as well as its production cost.

Let M denote the set of available raw materials

$$M = \{\text{SiO}_2, \text{Li}_2\text{O}, \text{Na}_2\text{O}, \text{K}_2\text{O}, \text{MgO}, \text{CaO}, \text{SrO}, \text{BaO}, \text{B}_2\text{O}_3, \text{Al}_2\text{O}_3\}.$$

For every material $i \in M$, $x_i \in [0, 1]$ denotes its mass fraction in the final glass. Furthermore, $y_i \in \{0, 1\}$ indicates whether material i is used. The unit cost of material i is denoted by c_i .

The customer specifies acceptable ranges for several material properties:

$$P = \{\text{conductivity, thermal expansion, refractive index, density, opacity}\}$$

For every property $k \in P$, a linear model predicts the value of property k given the glass composition:

$$P_k(x) = \sum_{i \in M} a_{ki} x_i.$$

Formulate the optimization problem mathematically using only linear constraints.



a) The objective is to minimize production cost.



b) The total composition must sum to 100%.



c) Silicon dioxide (SiO_2) forms the glass network. For mechanical stability, at least 60% of the glass must consist of SiO_2 .



d) Every selected material must contribute at least 2%.

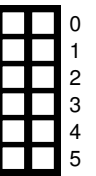


e) The produced glass needs to satisfy the property specifications defined by lower bounds L_k and upper bounds U_k for all properties.



f) Due to an existing patent, lithium oxide (Li_2O) and potassium oxide (K_2O) may not be used simultaneously.





g) Barium oxide (BaO) increases the refractive index of the glass but reduces its chemical durability. Therefore, barium oxide needs to be combined with aluminum oxide (Al_2O_3), to stabilize the glass network. Consequently, whenever BaO is used, Al_2O_3 must also be present.

h) At most six different raw materials may be used.

i) Given an integer variable $z \in \{0, 1\}$ and a variable $x \in [0, 1]$ model the product zx linearly using a new variable w .

j) *Boron anomaly:*

The coefficient of thermal expansion is increasing for small concentrations of boron oxide (B_2O_3), but changes significantly once the concentration exceeds a threshold τ . Consequently, the thermal expansion can no longer be described by a single linear function.

Incorporate this effect into the optimization model.

Hint: introduce a binary variable $z \in \{0, 1\}$ that captures whether the concentration of B_2O_3 is above, or below the threshold τ and enforce this behavior suitably using linear constraints.

Use the z to define the new thermal expansion model $P_{\text{exp}}(x, z)$.

You can use the previous subproblem without proof to eliminate products of z and x_i .





Problem 3 Duality (15 credits)

Consider the following primal linear program (P):

$$\begin{aligned} \max \quad & 4x_1 + 2x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 + 2x_3 \leq 8 \\ & 2x_1 - x_2 + x_3 \geq 2 \\ & x_1 + x_2 = 4 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

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a) Derive the dual linear program (D) corresponding to (P).





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b) State the primal and the dual complementary slackness conditions explicitly.

c) Consider the feasible primal solution

$$x^* = (2, 2, 0)^T$$

and the feasible dual solution

$$y^* = (2, 0, 2)^T.$$

Can these two solutions both be optimal? Justify your answer.

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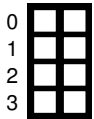


Problem 4 Branch and Bound (15 credits)

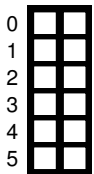
We consider the following knapsack problem with free variables:

$$\begin{aligned} \max_x \quad & \sum_i v_i x_i = 20x_1 + 12x_2 + 2x_3 + 9x_4 \\ \text{s.t.} \quad & \sum_i w_i x_i = 2x_1 + 2x_2 + x_3 + 2x_4 \leq c = 5. \end{aligned} \quad (\mathcal{P}_{\text{free}})$$

The above model is denoted by $\mathcal{P}_{\text{free}}$, in which the variables can take any negative or positive real value. We also consider the same problem when the variables are restricted to be non-negative reals, non-negative integers, or binary values, and denote them by $\mathcal{P}_{\geq 0}$, $\mathcal{P}_{\mathbb{Z}_+}$, and $\mathcal{P}_{\{0,1\}}$, respectively. We want to solve these problems using the Branch and Bound algorithm.



a) Are the problems $\mathcal{P}_{\text{free}}$ and $\mathcal{P}_{\geq 0}$ useful in solving the integer valued programs? Give your reasoning.



b) Apply the Branch and Bound algorithm for solving $\mathcal{P}_{\mathbb{Z}_+}$. Only iterate the first branch and solve the resulting subproblems. Also explain how the algorithm would further branch out without solving the resulting subproblems. You do not need to find the optimal solution. *Hint: The LP relaxation of $\mathcal{P}_{\mathbb{Z}_+}$ has the following optimal solution, which you may use without justification: Determine the value-weight ratios $q_i = v_i/w_i$ and index the items such that $q_j \geq q_{j+1}$ for all $j = 1, \dots, n-1$. Set $x_k = c/w_k$ for $k = 1$, and $x_j = 0$ for $j > 1$. When further constraints are imposed, the optimal solution greedily takes items in decreasing order of their value-to-weight ratio until the capacity is full.*





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c) Solve $\mathcal{P}_{\{0,1\}}$ using the Branch and Bound algorithm. *Hint: The LP relaxation of $\mathcal{P}_{\{0,1\}}$ has the following optimal solution, which you may use without justification: Determine the value-weight ratios $q_i = v_i/w_i$ and index the items such that $q_j \geq q_{j+1}$ for all $j = 1, \dots, n-1$. Choose k such that $\sum_{j=1}^k w_j \leq c$, but $\sum_{j=1}^{k+1} w_j > c$. Set $x_j = 1$ for $j = 1, \dots, k$, $x_{k+1} = (c - \sum_{j=1}^k w_j)/w_{k+1}$, and $x_j = 0$ for $j > k+1$. When further constraints are imposed, the optimal solution greedily takes items in decreasing order of their value-to-weight ratio until the capacity is full.*





Problem 5 Gomory Cut (15 credits)

Consider the following integer programme:

$$\begin{aligned} \max z &= 5x_1 + 8x_2 && (5.1) \\ \text{s.t. } x_1 + x_2 + s_1 &= 6 && (5.2) \\ 5x_1 + 9x_2 + s_2 &= 45 && (5.3) \\ s_1, s_2 &\geq 0 && (5.4) \\ x_1, x_2 &\in \mathbb{N}_0 && (5.5) \end{aligned}$$

After solving the LP relaxation with the simplex method, the following optimal tableau is obtained:

	x_1	x_2	s_1	s_2	RHS
z	0	0	$5/4$	$3/4$	$165/4$
x_1	1	0	$9/4$	$-1/4$	$9/4$
x_2	0	1	$-5/4$	$1/4$	$15/4$

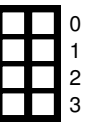
a) Determine the LP-optimal solution and explain why it is not feasible for the integer program.

b) Derive a Gomory fractional cut from the row corresponding to the basic variable x_2 . Show all intermediate calculations.





c) Verify that the current LP-optimal solution violates the Gomory cut obtained in the previous part.



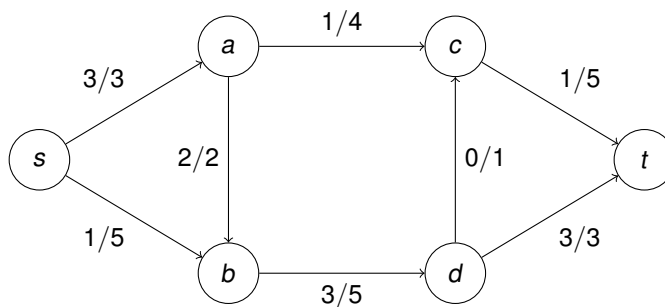
d) Formulate the LP relaxation that would be solved after adding the cut obtained in Part (2). You may reference original constraints (1)–(5) instead of rewriting them completely.



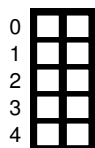


Problem 6 Maximum Flow (17 credits)

Consider the following flow network. The label f/u on an arc denotes its current flow f and capacity u .



a) State the value of the current flow.



b) Determine all arcs with positive capacity in the residual network and state their capacities. You may present your answer as a drawing or a list.





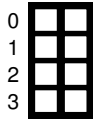
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c) Starting from the given flow, run the Ford–Fulkerson algorithm until no augmenting s - t path remains. For every augmentation, state:

- the chosen s - t path in the current residual network;
- the amount by which the flow is augmented; and
- the resulting flow on every original arc whose flow changes.

Finally, state the value of the resulting flow.





d) Find an s - t cut whose capacity equals the value of the resulting flow. State the two node sets defining the cut and compute its capacity. Explain why this proves that the resulting flow is maximum.



e) Briefly justify why a maximum flow can be chosen integral when all capacities are integral.





Problem 7 Budget Allocation (22 credits)

A company allocates one unit of budget between two advertising channels. Let $x_1 \in [0, 1]$ denote the fraction of the budget assigned to channel 1 and $x_2 \in [0, 1]$ the fraction assigned to channel 2. Due to diminishing returns, the resulting utility is modeled as

$$f_\alpha(x) = \log(1 + x_1) + \alpha \log(1 + x_2),$$

where $\alpha > 0$ describes the relative importance of channel 2 and $\log$ denotes the natural logarithm. The company wants to solve

$$(P_\alpha) \quad \max_{x \in \mathbb{R}^2} \log(1 + x_1) + \alpha \log(1 + x_2)$$

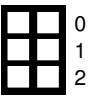
subject to

$$x_1 + x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0.$$

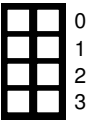
Equivalently, the inequality constraints can be written as

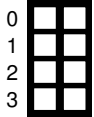
$$g_1(x) = x_1 + x_2 - 1 \leq 0, \quad g_2(x) = -x_1 \leq 0, \quad g_3(x) = -x_2 \leq 0.$$

a) State Slater's condition for a convex optimization problem with inequality constraints.

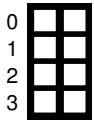


b) Show that f_α is concave on the feasible set.





c) Show that the feasible set of (P_α) is convex and that Slater's condition holds.



d) In addition to the utility, the company accounts for an administrative overhead proportional to the larger of the two allocations,

$$c(x) = \max\{x_1, x_2\}.$$

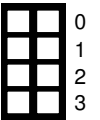
State the definition of a convex function and show *directly from the definition* that c is convex on $\mathbb{R}^2$.



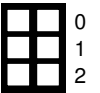


In the following subproblems, set up the KKT conditions of (P_α) and determine the optimal solution x^* for arbitrary $\alpha > 0$, distinguishing cases for α . Work through the following steps.

e) Write (P_α) as a minimization problem, state the Lagrangian, and write down all KKT conditions.



f) Show that the budget constraint is active at every KKT point, i.e. $x_1 + x_2 = 1$.





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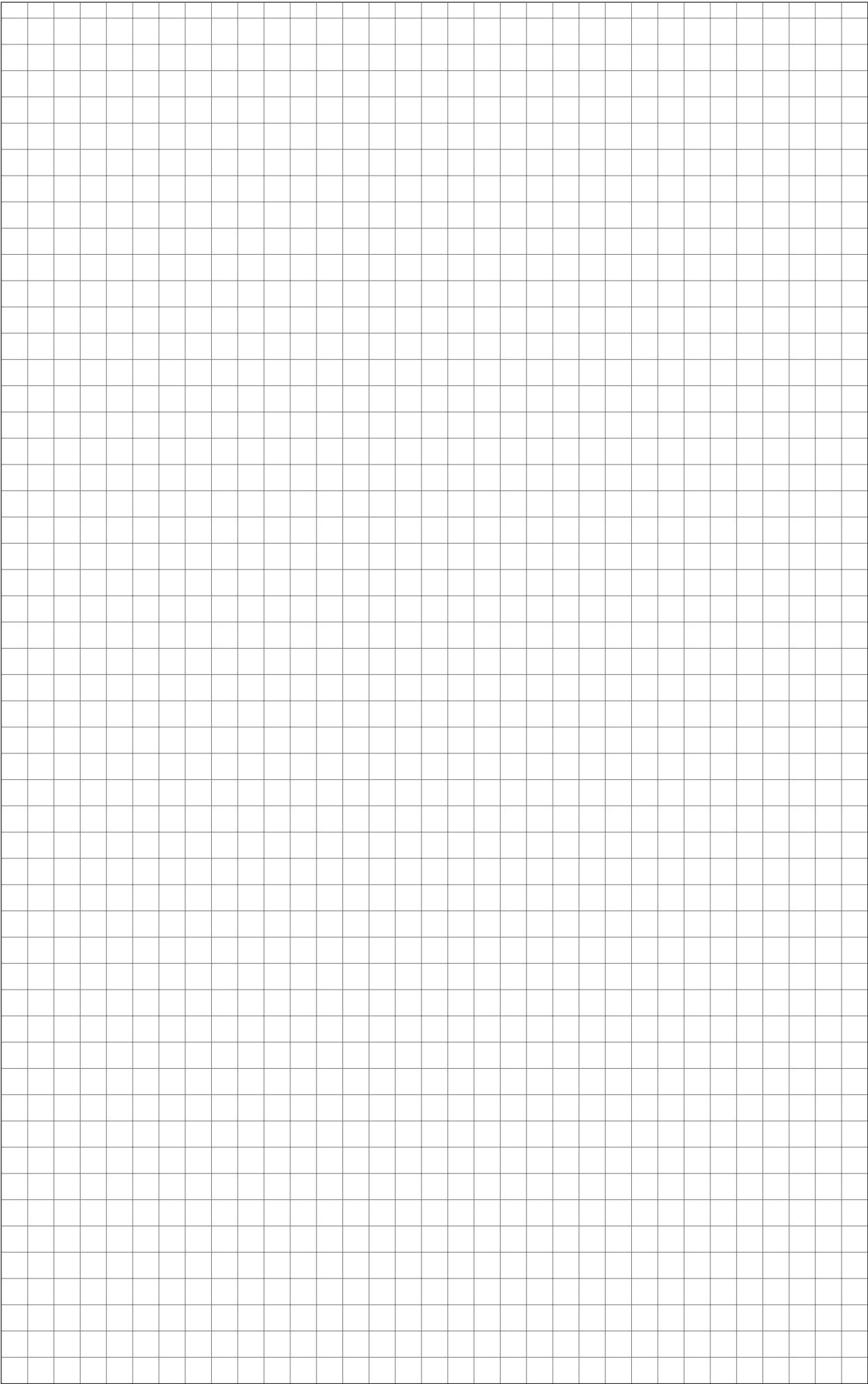
g) Using $x_2 = 1 - x_1$ from (f), analyse the three cases $x_1, x_2 > 0$, $x^* = (1, 0)^T$, and $x^* = (0, 1)^T$. For each case determine the multipliers and the range of α for which the case is consistent with the KKT conditions. Summarize the result and justify why the KKT points are global maxima.





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